(a)

Delay RMSE

Delay RMSE (samples)

10^{1}

10^{0}

10^{-1}

10^{-2}

−15 −10 −5 0 5 10 15

SNR (dB)

(b)

Magnitude RMSE

Magnitude RMSE

10^{-1}

10^{-2}

−15 −10 −5 0 5 10 15

SNR (dB)

(c)

Phase RMSE

Phase RMSE (rad)

10^{0}

10^{-1}

10^{-2}

−15 −10 −5 0 5 10 15

SNR (dB)

SBL baselines

Grid SBL

Off-grid SBL

Root SBL

CD-SAGE, restricted

Joint reference

(d)

Delay RMSE

Delay RMSE (samples)

10^{1}

10^{0}

10^{-1}

10^{-2}

SNR (dB)

15

−15 −10 −5 0 5 10 15

SNR (dB)

(e)

Magnitude RMSE

Magnitude RMSE

10^{-1}

10^{-2}

SNR (dB)

15

−15 −10 −5 0 5 10 15

SNR (dB)

(f)

Phase RMSE

Phase RMSE (rad)

10^{0}

10^{-1}

10^{-2}

SNR (dB)

15

−15 −10 −5 0 5 10 15

SNR (dB)

Refined MF/OMP baselines

MF + grid

OMP + grid

CD-SAGE, restricted

Joint reference